

Human Eye Correction Simulator

A Low-Cost Optical Model for Explaining Hyperopia and Its Correction

Prepared by: **Hammadi Nibel and MESSAHLI Hiba Manel**

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Project Idea

This project builds a large physical model of the human eye to show, in a visible way, how images form on the retina, why farsightedness (hyperopia) causes blur, and how a converging lens used in medical glasses restores clear vision.

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1 Introduction

The human eye is an amazing natural optical system. It works in a way that is similar to a camera: light enters the eye, the cornea and crystalline lens bend the light, and an image forms on the retina at the back of the eye.

However, if the geometry or optical power of the eye is not correct, the image does not form exactly on the retina. In that case, vision becomes blurry. One important example is **hyperopia** (farsightedness), where the eye does not converge light strongly enough, so the image tends to form *behind* the retina.

This project demonstrates that effect using a physical model eye made from low-cost materials. It also shows how a **converging lens** placed in front of the eye acts like corrective glasses and moves the image back onto the retina.

Why this project is interesting

Instead of only saying “glasses help people see,” this project lets the audience **watch the image move**. That makes the optics easy to understand and visually impressive.

2 Objectives

- To explain how the human eye forms images.
- To compare normal vision and hyperopia.
- To show how a converging lens corrects farsightedness.
- To build a realistic but low-cost classroom model.
- To connect biology, physics, and engineering in one demonstration.

3 How the Human Eye Forms Images

The eye is a converging optical system. Its main refracting parts are:

- the **cornea**, which provides much of the eye’s refractive power,
- the **crystalline lens**, which fine-tunes the focus,
- and the **retina**, which acts like a screen that receives the image.

When light rays from an object enter the eye, they are bent inward so that they meet at an image point on the retina. If the focus lands exactly on the retina, the image appears sharp.

Simple explanation

Think of the eye as a biological camera. The cornea and lens focus the light. The retina is the screen. If the focus is on the screen, the picture is sharp. If the focus is behind the screen, the picture becomes blurry.

The reference pictures of the eye all communicate the same physical message: the cornea and lens bend light, the retina receives the image, and the success or failure of vision depends on *where the rays finally meet*. To make that idea clearer for students, Figure 1 summarizes the main parts of the eye in a single classroom-style diagram.

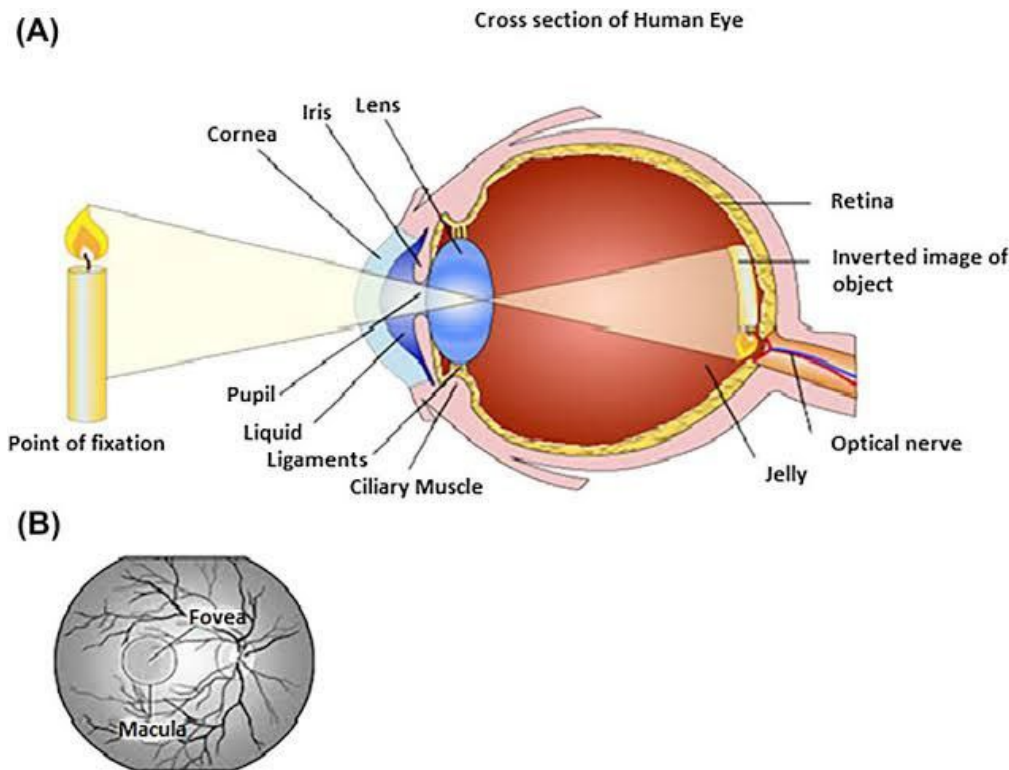


Figure 1: Cross-sectional image of the human eye showing the main optical structures involved in image formation. This real classroom image replaces the previous schematic drawing while preserving the document style.

How to read Figure 1

The **cornea** gives the first strong bending of light. The **lens** adjusts the focus. The **retina** is the light-sensitive layer that receives the image. The **fovea** is the region of sharpest vision. If the rays arrive on the retina, the image is clear; if they would meet only after the retina, the image on the retina is blurred.

4 Normal Vision vs Hyperopia

4.1 Normal Vision

In a normal eye, light from a distant object enters as nearly parallel rays. The optical power of the eye is just right, so these rays converge exactly on the retina.

That means:

- the image forms on the retina,
- the retina receives a sharp image,
- and the person sees clearly.

4.2 Hyperopia (Farsightedness)

In hyperopia, the eye is not converging the light enough. This usually happens because:

- the eyeball is too short, or
- the optical power of the cornea/lens system is too weak.

As a result, the focus point lies behind the retina.

That means:

- nearby objects become blurry,
- the retina intercepts the rays before they meet,
- and the image is not sharply formed on the retinal surface.

you may ask why a hyperopic person may still see distant objects reasonably well. The reason is that the eye can sometimes use **accommodation**: the ciliary muscles change the shape of the crystalline lens to increase the optical power. But this muscular effort cannot always compensate enough, especially for nearby objects. That is why reading or looking at close objects becomes difficult first.

Situation	Where the rays would meet	What the student should imagine
Normal eye	On the retina	The screen is exactly at the place where the image forms.
Hyperopic eye	Behind the retina	The screen is too early, so the rays have not finished meeting yet.
Corrected hyperopic eye	On the retina again	Glasses start the convergence earlier, so the eye finishes focusing at the correct place.

Key idea

Hyperopia is not mainly a “retina problem.” It is a **focusing problem**. The retina is in the wrong place relative to the focus.

5 How a Converging Lens Corrects Hyperopia

A converging lens, also called a convex lens, bends light inward before it enters the eye.

So instead of waiting until after the eye lens to become strongly convergent, the rays are already partially converging when they enter the eye. This increases the total optical power of the system.

Therefore:

- without glasses, the focus is behind the retina,
- with a converging lens, the focus shifts forward,
- and the image can form directly on the retina.

Friendly interpretation

The glasses do not “heal” the eye. They simply prepare the light better before it enters the eye, so the eye can finish the job correctly.

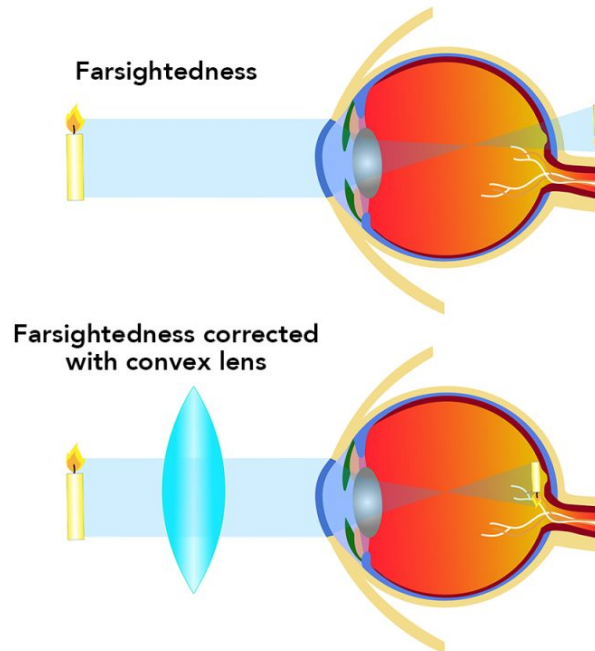


Figure 2: Reference image showing how Converging Lens Corrects Hyperopia.

6 Optical Theory

6.1 Thin Lens Formula

The main equation used in this project is the thin lens equation:

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} \quad (1)$$

where:

- d_o = object distance,
- d_i = image distance,
- f = focal length of the lens.

For a converging lens, $f > 0$.

6.2 Magnification

The magnification is:

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o} \quad (2)$$

where:

- h_o = object height,
- h_i = image height.

If $m < 0$, the image is inverted. If $|m| > 1$, the image is enlarged. If $|m| < 1$, the image is reduced.

6.3 Sample Calculation

Suppose the main eye lens has focal length:

$$f = 12 \text{ cm}$$

and the object is placed at:

$$d_o = 36 \text{ cm}$$

Then:

$$\begin{aligned} \frac{1}{d_i} &= \frac{1}{12} - \frac{1}{36} \\ \frac{1}{d_i} &= \frac{3-1}{36} = \frac{2}{36} = \frac{1}{18} \end{aligned}$$

So:

$$d_i = 18 \text{ cm}$$

This means the image forms 18 cm behind the lens.

Now the interpretation:

- If the retina is at 18 cm, vision is normal.
- If the retina is at 16 cm, the focus is behind the retina, which simulates hyperopia.
- If we add a converging corrective lens, the image can be shifted forward again to 16 cm.

6.4 Why Near Objects Are More Difficult

Equation (1) explains an important classroom observation. When the object gets closer, d_o becomes smaller, so $1/d_o$ becomes larger. To keep the lens equation true, the image position must adjust. The eye therefore needs extra optical power to keep the image on the retina.

For a healthy eye, accommodation provides part of that extra power. For a hyperopic eye, the system already starts with insufficient power, so close objects become problematic sooner.

Student interpretation of the equation

When the object comes closer, the eye has to work harder. In a hyperopic eye, the focusing system is already slightly too weak, so the image moves behind the retina even more easily for near objects.

6.5 Graphical Explanation of Image Position

The graph in Figure 3 converts the thin-lens formula into a visual narrative. For a model-eye lens with focal length $f = 12$ cm, the blue curve traces how the image distance d_i depends on the object distance d_o . Two horizontal dashed lines mark proposed retina positions: 18 cm for a normal model eye and 16 cm for a hyperopic model eye. Whenever the curve rises *above* a retina line, the image forms behind that retina, producing a blurry picture. The shaded regions in the graph make those “blur zones” immediately visible, so students can compare how much of the object-distance range is affected in each case.

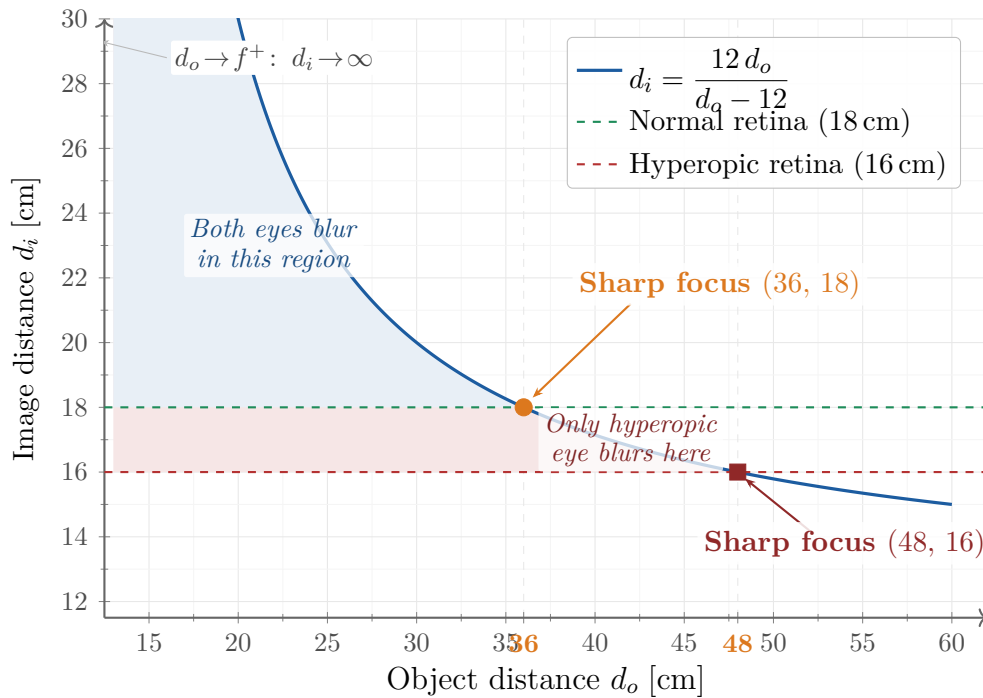


Figure 3: Image distance d_i vs. object distance d_o for $f = 12$ cm. Light-red zone ($13 < d_o < 48$ cm): image forms behind the hyperopic retina. Light-blue zone ($13 < d_o < 36$ cm): even the normal retina receives a blurred image. Orange dot: normal-eye crossing (36, 18); red square: hyperopic-eye crossing (48, 16).

From Figure 3, students can read the key conclusions directly off the graph:

- For the **hyperopic retina** at 16 cm, the blue curve stays above the red dashed line from $d_o = 13$ all the way to $d_o = 48$ cm. That entire light-red region represents blurred vision.
- For the **normal retina** at 18 cm, only the range $d_o < 36$ cm causes blur — a noticeably smaller zone.
- The hyperopic eye therefore loses sharp focus over a *much wider* range of object distances, which is exactly why farsighted people struggle with more everyday tasks.
- The further left on the graph (closer objects), the higher the curve rises, meaning the focus drifts even further behind the retina and the blur worsens.
- Near the asymptote at $d_o = f = 12$ cm, the image distance diverges to infinity: placing an object at the focal plane produces no real image at all.

6.6 Graphical Explanation of Corrective Power

Another helpful graph is to compare the optical power of the system. Optical power is measured in diopters and is defined by

$$P = \frac{1}{f [\text{m}]} \quad (3)$$

where f is expressed in meters. A converging corrective lens adds positive power to the eye.

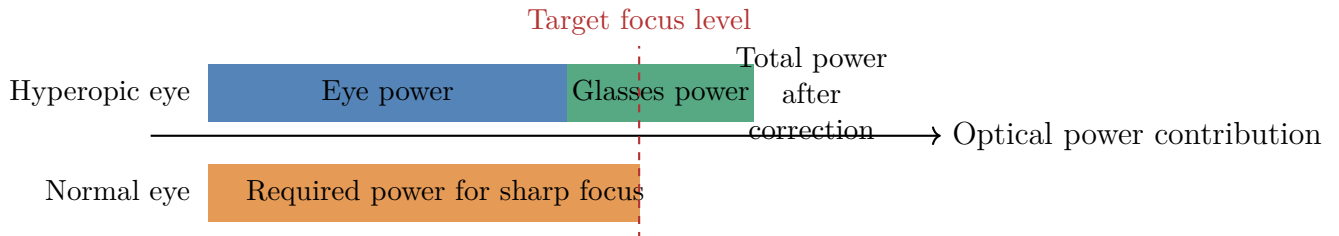


Figure 4: Qualitative power diagram. The hyperopic eye alone provides too little focusing power, while the glasses add the missing amount so the total reaches the required level.

Figure 4 is not a medical prescription graph. It is a teaching graph. Its purpose is to show that the glasses do not replace the eye; they *add the missing optical power* so that the combined system can focus on the retina.

7 Ray Diagram Explanation

Below are three simplified ray diagrams. They correspond closely to the reference pictures: one diagram for normal vision, one for hyperopia where the focus lies behind the retina, and one for correction with a converging lens placed in front of the eye.

7.1 Normal Eye

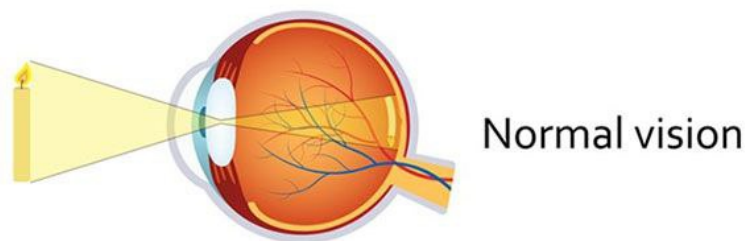


Figure 5: Normal vision: the top panel of the provided eye diagram shows rays converging correctly on the retina.

In this first case, the eye lens supplies exactly the right convergence. The retina and the focus location coincide. Students should notice that the orange rays meet *exactly at* the retinal line and not before or after it.

7.2 Hyperopic Eye

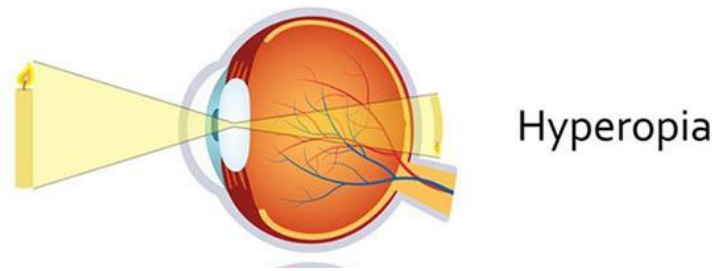


Figure 6: Hyperopia: the middle panel of the provided diagram shows the focus forming behind the retina, so the retinal image is blurred.

In the hyperopic case, the rays are still moving toward one another when they strike the retina. Therefore, the retinal screen is illuminated by a small blurred region rather than a single sharp point. This is the central idea of farsightedness in this project.

7.3 Corrected Hyperopic Eye

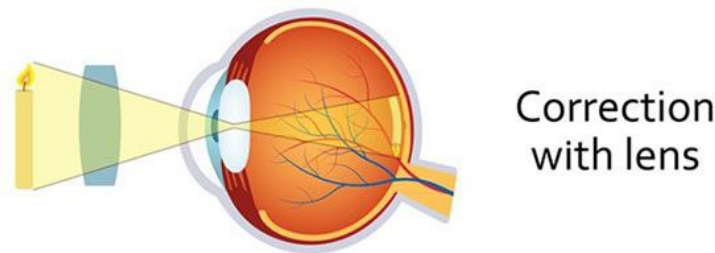


Figure 7: Correction of hyperopia using a converging lens in front of the eye, shown in the bottom panel of the provided diagram.

With the corrective lens, the rays enter the eye already partially converging. Then the eye lens completes the focusing process so that the meeting point comes back onto the retina.

8 Additional Scientific Notes for Students

8.1 Why the Image in the Eye Is Inverted

The lens of the eye forms a real image on the retina, and real images formed by a converging lens are generally inverted. This is not a problem for perception because the brain interprets the retinal signals and constructs a stable view of the world.

8.2 Why the Fovea Matters

The fovea is the region of the retina with the highest visual acuity. In everyday vision, the eye naturally tries to place important details on the fovea because it gives the sharpest possible image.

9 Conclusion

This project shows that hyperopia is a focusing problem in which the image forms behind the retina. A converging lens corrects this by increasing the optical power of the system and shifting the focus forward onto the retina.

Because the image movement can be seen directly on the tracing-paper retina, the demonstration is not only scientifically correct but also visually strong. It combines medicine, optics, physics, and engineering in one clear and low-cost experiment.